AR40215

# Building Energy System and Auditing

Semester-VI, B.Arch

## Building Energy Index & Auditing
## Life Cycle Analysis

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Life cycle energy analysis is an approach that accounts for all energy inputs to a building in its whole life. The system boundaries of this analysis include the use of energy in the following three phases:

**(i) Construction,**
**(ii) Operation, and**
**(iii) Demolition.**

The Life Cycle Energy (LCE) of any building is the arithmetic sum of

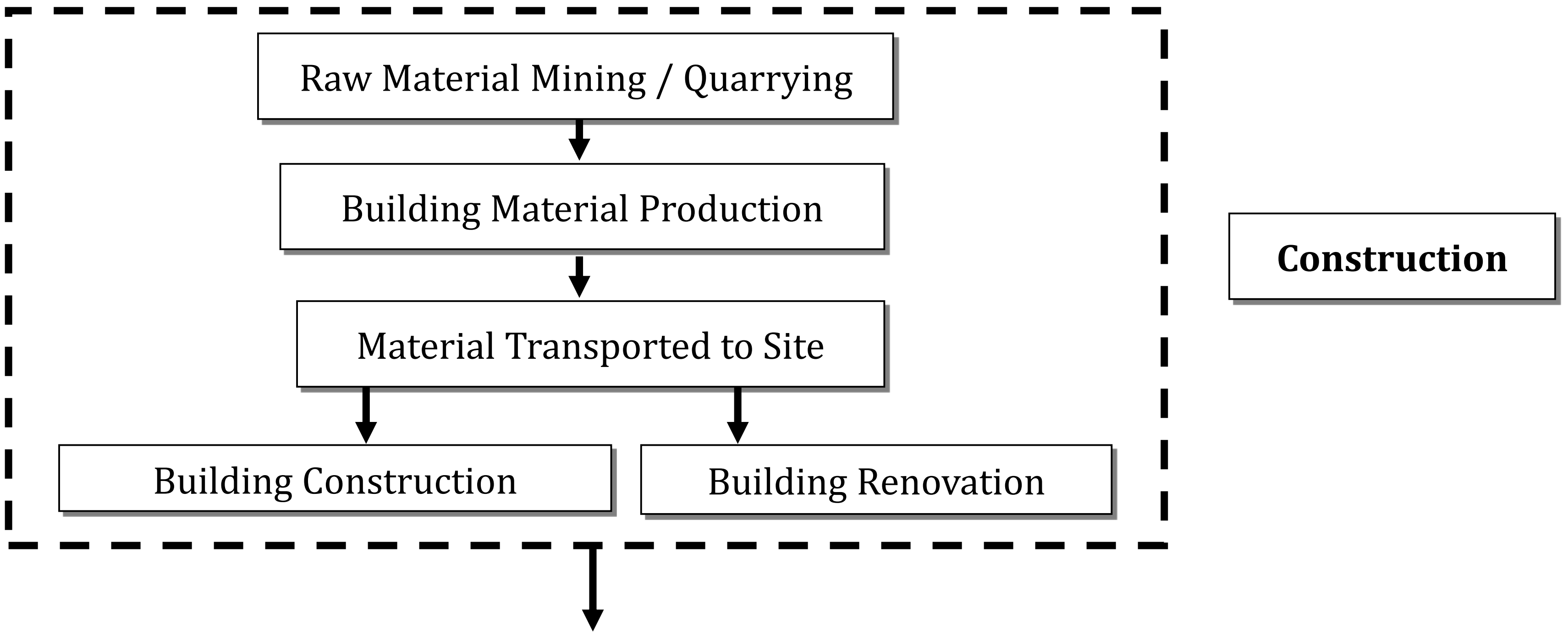
- **Embodied energy (EE),**
- **Operational energy (OE) and**
- **Demolition energy (DE).**

It can be represented mathematically as

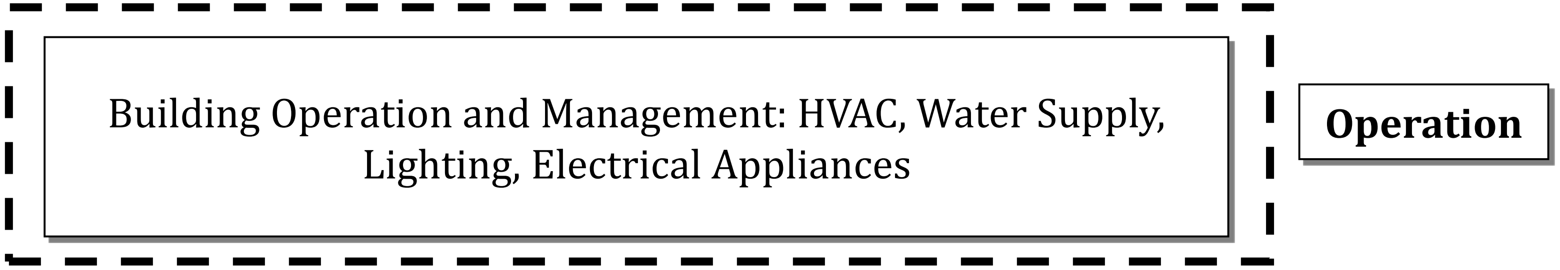
**LCE = EE + OE + DE**

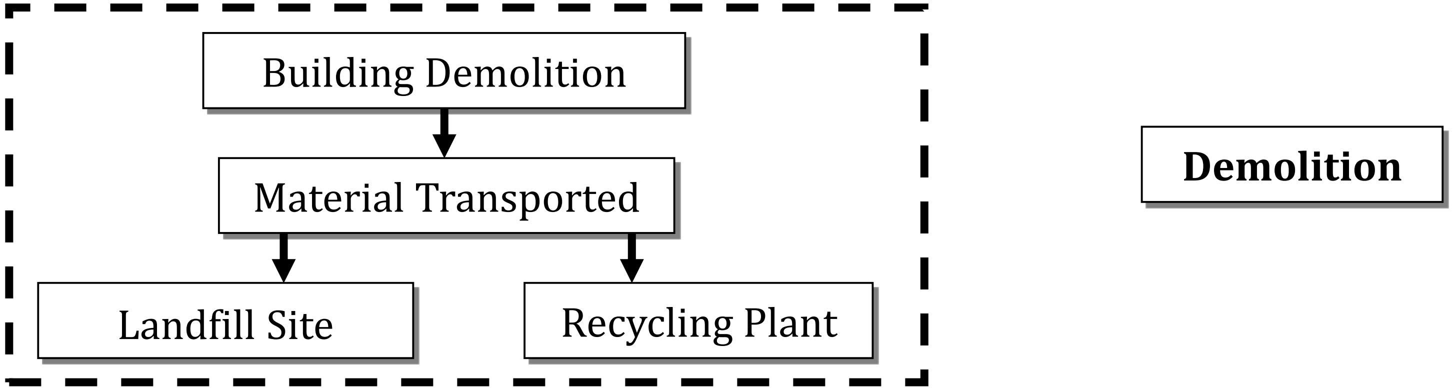
It is expressed in Kilo Watt – Hour (kWh) unit in absolute terms, and kWh/$m^2$ [dividing the absolute energy by the built-up area of the building] in relative terms

The energy required in the **Construction Phase** is also known as **Embodied Energy**, includes manufacturing and transportation of building materials and technical installations used in erection and renovation of the buildings.



**Operation Phase** encompasses all activities related to the use of the buildings, over its life span. These activities include maintaining comfort condition inside the buildings, water use and powering appliances




Finally, **Demolition Phase** includes destruction of the building and transportation of dismantled materials to landfill sites and/or recycling plants. Energy use in each phase is discussed below.

## Embodied Energy (EE):

Embodied energy is the energy utilized during construction phase of the building.
It is the energy content of all the materials used in the building and technical installations, and energy incurred at the time of erection/construction and renovation of the building.
Energy content of materials refers to the energy used to acquire raw materials (excavation), manufacture and transport to the building site.
Embodied energy is divided in two parts:

**(i)** **Initial embodied energy** ($EE_I$) and
**(ii)** **Recurring embodied energy** ($EE_R$).

### Initial Embodied Energy ($EE_I$)

Initial embodied energy of a building is the energy incurred for initial construction of the building. It is the combination of the energy required for

**(i)** **Building Material Manufacturing,**
**(ii)** **Transportation to Site and**
**(iii)** **Construction and Erection at Site**

$$\boxed{EE_I = E_{Manuf} + E_{Trans\text{-}site} + E_{Cons}}$$

The energy requirement for producing all the building materials, $\mathbf{E_{Manuf}}$ is estimated as follows:

$$E_{Manuf} = \sum_{i=1}^{n} m_i \left(1 + \frac{w_i}{100}\right) \times M_i$$

Where,
n = total number of materials,
*i* = the material of concern,
$m_i$ = amount of the building material *i* (ton),
$w_i$ = the factor for waste of the material *i* produced during construction of the building expressed in percentage and
$M_i$ = energy required for manufacturing the building material *i* (kWh/ton).

The energy use, during transporting the building materials to the building site can be estimated as follows:

$$E_{Trans-site} = \sum_{i=1}^{n} m_i \left(1 + \frac{w_i}{100}\right) \times d_i \times T_c$$

Where, $d_i$ = distance from the manufacturer of material i to the building site (km) and
$T_c$ = energy required for the conveyance concerned (kWh/ton km)

The energy required for various construction processes can be obtained and the following equation can be use to determine the energy consumption during construction.

$$E_{Cons} = \sum_{j=1}^{m} p_j \times P_j$$

Where,
m = total number of processes,
j = the type of process under consideration,
$p_j$ = the amount of the process j expressed in ton, $m^3$ or $m^2$ usable floor area, and
$P_j$ = energy required for the process j (kWh/ton, kWh/$m^3$ or kWh/$m^2$ usable floor area.

$$\mathbf{EE_I = E_{Manuf} + E_{Trans\text{-}site} + E_{Cons}}$$

## Recurring Embodied Energy ($EE_R$)

Some of the building materials may have a **life span less than that of the building**. As a result, they are replaced to renovate/rehabilitate the building.
In addition to this, buildings require some **regular annual maintenance**.
The energy incurred for such **repair, renovate and replacement** needs to be accounted during the entire life of the buildings.

The sum of the energy embodied in the material, used in the rehabilitation and maintenance is called recurring embodied energy and can be expressed as:

$$\mathbf{EE_R = E_{Manuf\text{-}renov} + E_{Trans\text{-}renov} + E_{Cons\text{-}renov}}$$

The energy use for producing the building materials during the renovation, is estimated as:

$$E_{Manuf-renov} = \sum_{i=1}^{n} m_i \left(1 + \frac{w_i}{100}\right) \times M_i \times \left(\frac{\text{Life Span of a building}}{\text{Life span of material } i} - 1\right)$$

The energy required for transportation of renovation material from manufacturing unit to site and transportation of out of date material from site to disposal site can be calculated as:

$$E_{Trans-renov} = \sum_{i=1}^{n} m_i \left(1 + \frac{w_i}{100}\right) \times \left(\frac{\text{Life Span of a building}}{\text{Life span of material } i} - 1\right) \times (d_i + 20) \times T_c$$

*Here, the distance from the building site to the waste disposal site is assumed as 20 km*

The energy consumption during construction during renovation can be calculated exactly as it was given in case of building construction.

$$E_{Cons-renov} = \sum_{j=1}^{m} p_j \times P_j$$

## Operating energy (OE)

It is the energy required for **maintaining comfort conditions** and day-to-day maintenance of the buildings.

It is the energy for HVAC (heating, ventilation and air conditioning), domestic hot water, lighting, and for running appliances. Operational energy largely varies on the **level of comfort** required, **climatic conditions** and operating schedules.

**OE = Energy Consumption per year (in kWh) X Life span of building**

## Demolition Energy (DE)

At the end of buildings' service life, energy is required to **demolish the building and transporting the waste** material to landfill sites and/or recycling plants. This energy is termed as demolition energy and expressed as:

$$\mathbf{DE = E_{Demo} + E_{Trans\text{-}demo}}$$

During the demolition, some energy will also be needed for **different processes** and can be calculated as:

$$E_{Demo} = \sum_{j=1}^{m} p_j \times P_j$$

The **energy required for transportation** of demolished material from site to disposal site can be calculated as

$$E_{Trans-demo} = \sum_{i=1}^{n} m_i \left(1 + \frac{w_i}{100}\right) \times 20 \times T_c$$

# Thank You